

TABLE I. TEMPERATURE SENSOR DATA COMPARISON TABLE

| <b>Trial</b> | <b>Reference Measurement (°C)</b> | <b>System Measurement (°C)</b> | <b>Error (°C)</b> |
|--------------|-----------------------------------|--------------------------------|-------------------|
| 1            | 7.2                               | 7.67                           | 0.47              |
| 2            | 7.3                               | 7.64                           | 0.34              |
| 3            | 7.4                               | 7.73                           | 0.33              |
| 4            | 7.5                               | 7.87                           | 0.37              |
| 5            | 7.7                               | 8                              | 0.3               |
| 6            | 29.4                              | 28.81                          | 0.59              |
| 7            | 29.2                              | 28.81                          | 0.39              |
| 8            | 29.1                              | 28.81                          | 0.29              |
| 9            | 29.2                              | 28.75                          | 0.45              |
| 10           | 29.1                              | 28.75                          | 0.35              |
| 11           | 29.3                              | 28.87                          | 0.43              |
| 12           | 29.4                              | 28.81                          | 0.59              |
| 13           | 29.2                              | 28.81                          | 0.39              |
| 14           | 29.1                              | 28.81                          | 0.29              |
| 15           | 29.2                              | 28.75                          | 0.45              |

TABLE II. PH SENSOR DATA COMPARISON TABLE

| <b>Trial</b> | <b>pH Reference</b> | <b>pH Sensor System Measurement</b> | <b>Error</b> |
|--------------|---------------------|-------------------------------------|--------------|
| 1            | 6.86                | 6.89                                | 0.03         |
| 2            | 6.86                | 6.79                                | 0.07         |
| 3            | 6.86                | 6.89                                | 0.03         |
| 4            | 6.86                | 6.97                                | 0.11         |
| 5            | 6.86                | 6.9                                 | 0.04         |
| 6            | 6.86                | 7.03                                | 0.17         |
| 7            | 6.86                | 6.94                                | 0.08         |
| 8            | 6.86                | 7.02                                | 0.16         |
| 9            | 9.18                | 9.12                                | 0.06         |
| 10           | 9.18                | 9.3                                 | 0.12         |
| 11           | 9.18                | 9.23                                | 0.05         |
| 12           | 9.18                | 9.22                                | 0.04         |
| 13           | 9.18                | 9.24                                | 0.06         |
| 14           | 9.18                | 9.27                                | 0.09         |
| 15           | 9.18                | 9.21                                | 0.03         |

TABLE III. AVERAGE ERROR OF TEMPERATURE AND PH SENSORS

| Data             | Average Error  |
|------------------|----------------|
| Temperature cold | 0.385±0.044 °C |
| Temperature warm | 0.42±0.138 °C  |
| pH @ 6.86        | 0.143±0.087    |
| pH @ 9.18        | 0.063±0.039    |

TABLE IV. COMPARISON OF SCHEDULED AND ACTUAL FEEDING TIME

| Trial | Set Time              | Logged Dispense Time     | Delay (s) |
|-------|-----------------------|--------------------------|-----------|
| 1     | 2025-12-12<br>“01:35” | 2025-12-12<br>“01:35:05” | 5         |
| 2     | 2025-12-12<br>“01:36” | 2025-12-12<br>“01:36:03” | 3         |
| 3     | 2025-12-12<br>“01:40” | 2025-12-12<br>“01:40:03” | 3         |
| 4     | 2025-12-12<br>“01:41” | 2025-12-12<br>“01:41:05” | 5         |
| 5     | 2025-12-12<br>“01:42” | 2025-12-12<br>“01:42:05” | 5         |
| 6     | 2025-12-12<br>“01:43” | 2025-12-12<br>“01:43:03” | 3         |
| 7     | 2025-12-12<br>“01:44” | 2025-12-12<br>“01:44:05” | 5         |
| 8     | 2025-12-12<br>“01:45” | 2025-12-12<br>“01:45:05” | 5         |
| 9     | 2025-12-12<br>“01:46” | 2025-12-12<br>“01:46:03” | 3         |
| 10    | 2025-12-12<br>“01:47” | 2025-12-12<br>“01:47:03” | 3         |
| 11    | 2025-12-24<br>“15:37” | 2025-12-24<br>“15:37:03” | 3         |
| 12    | 2025-12-24<br>“15:40” | 2025-12-24<br>“15:40:03” | 3         |
| 13    | 2025-12-24<br>“15:41” | 2025-12-24<br>“15:41:05” | 5         |
| 14    | 2025-12-24<br>“15:42” | 2025-12-24<br>“15:42:04” | 4         |
| 15    | 2025-12-24<br>“15:43” | 2025-12-24<br>“15:43:04” | 4         |
| 16    | 2025-12-24<br>“15:45” | 2025-12-24<br>“15:45:04” | 4         |
| 17    | 2025-12-24<br>“15:46” | 2025-12-24<br>“15:46:03” | 3         |
| 18    | 2025-12-24<br>“15:47” | 2025-12-24<br>“15:47:03” | 3         |
| 19    | 2025-12-24<br>“15:48” | 2025-12-24<br>“15:48:03” | 3         |
| 20    | 2025-12-24<br>“15:49” | 2025-12-24<br>“15:49:04” | 4         |
| 21    | 2025-12-24<br>“15:50” | 2025-12-24<br>“15:50:04” | 4         |
| 22    | 2025-12-25<br>“21:51” | 2025-12-25<br>“21:51:04” | 4         |
| 23    | 2025-12-25<br>“21:54” | 2025-12-25<br>”21:54:03” | 3         |
| 24    | 2025-12-25<br>“21:55” | 2025-12-25<br>”21:55:04” | 4         |
| 25    | 2025-12-25<br>“21:56” | 2025-12-25<br>”21:56:04” | 4         |
| 26    | 2025-12-25<br>“21:57” | 2025-12-25<br>”21:57:04” | 4         |
| 27    | 2025-12-25<br>“21:58” | 2025-12-25<br>”21:58:03” | 3         |
| 28    | 25-12-25<br>“21:59”   | 2025-12-25<br>”21:59:05” | 5         |
| 29    | 25-12-25<br>“22:00”   | 2025-12-25<br>”22:00:04  | 4         |

|    |                     |                          |   |
|----|---------------------|--------------------------|---|
| 30 | 25-12-25<br>"22:01" | 2025-12-25<br>"22:01:03" | 3 |
|----|---------------------|--------------------------|---|

TABLE V. FEEDING PORTION ACCURACY

| Target Weight<br>(gram) | Average Error<br>(gram) | Percentage of<br>Target |
|-------------------------|-------------------------|-------------------------|
| 0.1                     | 0.004±0.003             | 4%                      |
| 0.2                     | 0.012±0.005             | 6%                      |
| 0.3                     | 0.012±0.006             | 4%                      |
| 0.4                     | 0.012±0.004             | 3%                      |